Exponents & Scientific Notation: Mini-Lessons

NC.8.EE.1, 3–4

ANSWER KEY

2.1 What an exponent means NC.8.EE.1

An **exponent** is a shortcut for multiplying the same number over and over. $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.

Base
the number being multiplied (in 2^5, the 2)

Exponent
how many times the base is multiplied (the 5)

Expanded form
written out the long way: 2 × 2 × 2 × 2 × 2

base 2^{5} exponent

$$2 \times 2 \times 2 \times 2 \times 2 = 32$$

NOT 2 × 5 = 10

Parentheses matter: $(-3)^2 = (-3)(-3) = 9$, but $-3^2 = -(3 \times 3) = -9$.

Watch out: 2^5 is **not** 2 × 5. Write it out to check.

Walk through: Find $(-2)^3$

1. Write $(-2)^3$ in expanded form.
   $(-2)(-2)(-2) \rightarrow (-2)(-2)(-2)$
2. Multiply the first two: $(-2)(-2) = ?$
   $4 \rightarrow = 4 \cdot (-2)$
3. Now $4 \cdot (-2) = ?$
   $-8 \rightarrow = -8$

Check your understanding

1 Score 3 NC.8.EE.1

Find the value of 3^4.

Answer: **81**

2 Score 3 NC.8.EE.1

Which is the same as 5^3?

(A) 5 + 5 + 5
(B) 5 × 3
(C) 3 × 3 × 3 × 3 × 3
(D) **5 × 5 × 5**

3 Score 3 NC.8.EE.1

Find the value of $(-4)^2$.

Answer: **16**

2.2 Multiply and divide powers NC.8.EE.1

Same base: when you **multiply**, **add** the exponents. When you **divide**, **subtract** the exponents. $x^a \cdot x^b = x^{a+b}$ and $x^a \div x^b = x^{a-b}$.

Same base
the numbers or letters being raised to powers match: 2^3 and 2^4

Coefficient
the number in front: in $3x^2$, the 3

$$x^3 \cdot x^4$$

$$(x \cdot x \cdot x) \cdot (x \cdot x \cdot x \cdot x)$$

7 x's multiplied

$$x^3 \cdot x^4 = x^{3+4} = x^7$$

Multiply same base → ADD the exponents

$2^3 \times 2^4 = 2^{3+4} = 2^7$

$x^5 \div x^2 = x^{5-2} = x^3$

$(3x^2)(2x^4) = 6x^6$ (multiply the numbers, add the exponents)

Watch out: Do **not** multiply the bases. $2^3 \times 2^4$ is 2^7, not 4^7.

Walk through: Simplify $(4x^3)(2x^5)$

1. First, the numbers in front. 4 × 2 = ?
   8 → 4 × 2 = 8
2. Next, the x's: $x^3 \cdot x^5$. What do we do with the exponents?
   Add them: 3 + 5 = 8 → $x^3 \cdot x^5 = x^8$
3. Put it together.
   $8x^8$ → $(4x^3)(2x^5) = 8x^8$

Check your understanding

1 Score 3 NC.8.EE.1

Simplify: $5^2 \times 5^5$

(A) 5^{10} (B) 5^7 (C) 5^3 (D) 25^7

2 Score 3 NC.8.EE.1

Simplify: $a^7 \div a^3$

(A) a^4 (B) $a^{7/3}$ (C) a^{21} (D) a^{10}

2.3 Power of a power, zero and negative exponents NC.8.EE.1

Power of a power: multiply the exponents, $(x^3)^2 = x^6$. **Zero exponent:** any nonzero number to the 0 power is **1**. **Negative exponent:** flip it to a fraction, $2^{-3} = 1/2^3 = 1/8$.

Reciprocal
the flip of a number: the reciprocal of 5 is 1/5

Power of a power
a power raised to another power, like $(x^3)^2$

Rule	Example
$(x^a)^b = x^{a \cdot b}$	$(3^2)^3 = 3^6$
$x^0 = 1$	$9^0 = 1$
$x^{-a} = 1/x^a$	$10^{-2} = 1/100$

$2^3 = 8$
$2^2 = 4$ ÷ 2
$2^1 = 2$ ÷ 2
$2^0 = 1$ ÷ 2
$2^{-1} = 1/2$ ÷ 2
$2^{-2} = 1/4$ ÷ 2

Watch out: A negative exponent does **not** make the number negative. $2^{-3} = 1/8$, not -8.

Walk through: Write 5^{-2} as a fraction

1. The exponent is negative. What does that tell us?
   Flip it: $5^{-2} = 1/5^2 \rightarrow 5^{-2} = 1/5^2$
2. What is 5^2?
   $25 \rightarrow = 1/25$

Check your understanding

1 Score 3 NC.8.EE.1

Simplify: $(x^4)^2$

(A) x^2 (B) x^8 (C) x^6 (D) x^{16}

2 Score 3 NC.8.EE.1

Find the value of 9^0.

Answer: 1

3 Score 3 NC.8.EE.1

Find the value of 2^{-3}. (Type a fraction like 1/4.)

Answer: 1/8

2.4 Scientific notation NC.8.EE.3

Scientific notation writes a number as $a \times 10^n$, where **a is at least 1 and less than 10**. Big numbers get a **positive** exponent. Small numbers (less than 1) get a **negative** exponent.

Front number (a)

must be 1 or more, and less than 10: 4.5 ✓, 45 ×, 0.45 ×

Power of 10

the exponent counts how many places the decimal point moves



$0.00032 = 3.2 \times 10^{-4}$

front number 4.5 is between 1 and 10 ✓

Standard form	Hops	Scientific notation
45,000,000	7 left	4.5×10^7
0.00032	4 right	3.2×10^{-4}
6,020,000	6 left	6.02×10^6

Watch out: 45×10^3 is not finished: 45 is not less than 10. Fix it to 4.5×10^4.

Walk through: `Write 0.0061 in scientific notation`

1. Where should the decimal point go so the front number is between 1 and 10?
   After the 6: 6.1 → 6.1
2. How many places did the point hop?
   3 places to the right → 3 hops right
3. Positive or negative exponent?
   Negative: $6.1 \times 10^{-3} \rightarrow 0.0061 = 6.1 \times 10^{-3}$

Check your understanding

1 Score 3 NC.8.EE.3

Write `32,000` in scientific notation.

Ⓐ 3.2×10^{-4} Ⓑ 3.2×10^{3}

Ⓒ **3.2×10^{4}** Ⓓ 32×10^{3}

2 Score 3 NC.8.EE.3

Write 5×10^{3} in standard form.

Answer: **5000**

2.5 Multiply and divide in scientific notation NC.8.EE.4

Multiply: multiply the front numbers and **add** the exponents. **Divide:** divide the front numbers and **subtract** the exponents. Then check that the front number is between 1 and 10.

Adjust
if the front number is 10 or more, move the point and fix the exponent: $15 \times 10^{7} = 1.5 \times 10^{8}$

$$(8 \times 10^{7}) \div (4 \times 10^{3}) = (8 \div 4) \times 10^{7-3} = 2 \times 10^{4}$$

$$(3 \times 10^{4})(2 \times 10^{5})$$
$$= (3 \times 2) \times 10^{4+5}$$
$$= 6 \times 10^{9}$$

front numbers: multiply · tens: add exponents

Watch out: After multiplying, check the front number. 15×10^{7} must be fixed to 1.5×10^{8}.

Walk through: `Multiply (5 × 10³)(3 × 10⁴)`

1. Multiply the front numbers.
   $5 \times 3 = 15 \rightarrow 15$
2. Now the powers of 10: $10^{3} \cdot 10^{4} = ?$
   10^{7} (add 3 + 4) → 15×10^{7}
3. 15 is not less than 10. Fix it.
   $1.5 \times 10^{8} \rightarrow = 1.5 \times 10^{8}$

Check your understanding

1 Score 3 NC.8.EE.4

Multiply: $(2 \times 10^3)(4 \times 10^2)$

(A) 8×10^6

(B) 8×10^1

(C) **8×10^5**

(D) 6×10^5

2 Score 3 NC.8.EE.4

Divide: $(9 \times 10^8) \div (3 \times 10^2)$

(A) **3×10^6**

(B) 3×10^4

(C) 6×10^6

(D) 3×10^{10}